Lab1 - Experimenting with TCP using ns-3 Simulator

Total points for this lab (9 points + 1 bonus point)

In this lab, you will experiment with simulating the TCP protocol written in C++ using the ns-3 simulator. The ns-3 is a discrete-event network simulator for research and educational use. For this lab, no underlying knowledge about the simulator is required; but it is recommended that you study the online ns-3 tutorial at https://www.nsnam.org/docs/release/3.31/tutorial/html/index.html and https://www.nsnam.org/docs/release/3.31/models/html/tcp.html

You need to modify some of the parameters (such as link delay, channel loss probability) in the C++ code, run the TCP example, and graph the output. The output is the size of the congestion window over time. The objective is to study the impact of channel properties on the performance of TCP protocol performance. According to the results you get, you will answer the questions at the end of this lab.

Assuming ns-3 is installed and ready to run, follow the steps below to run one of the examples that is available inside the ns-3 directory.

- In ns-3, any example you run should be copied to the **scratch** directory before you run it. The scratch directory: /ns-allinone-3.30.1/ns-3.30.1/scratch.
- Copy the file /ns-allinone-3.30.1/ns-3.30.1/examples/tutorial/seventh.cc to the **scratch** directory.
- Run the example using the command below (do not include .cc):

  ```
  $ ./waf --run scratch/seventh
  ```

- The above example will build a network of two nodes and simulate it given the instructions inside **seventh.cc** example. Read the ns-3 tutorial and understand how the program seventh.cc works.

```
     node 0                  node 1
+----------------+    +----------------+
|   ns-3 TCP     |    |   ns-3 TCP     |
+----------------+    +----------------+
|    10.1.1.1    |    |    10.1.1.2    |
+----------------+    +----------------+
| point-to-point |    | point-to-point |
+----------------+    +----------------+
         |                     |
         +---------------------+
```

5 Mbps, 2 ms

- Information about the time and the size of the congestion window will be stored inside the file /ns-allinone-3.30.1/ns-3.30.1/seventh.cwnd. The file has three columns. The first column is the current time in seconds. The second and third columns are the old and the new value of the congestion window, respectively.
- To draw the congestion window, install the **Gnuplot** graphing software, if not installed.

```
$ sudo apt install gnuplot-nox
```

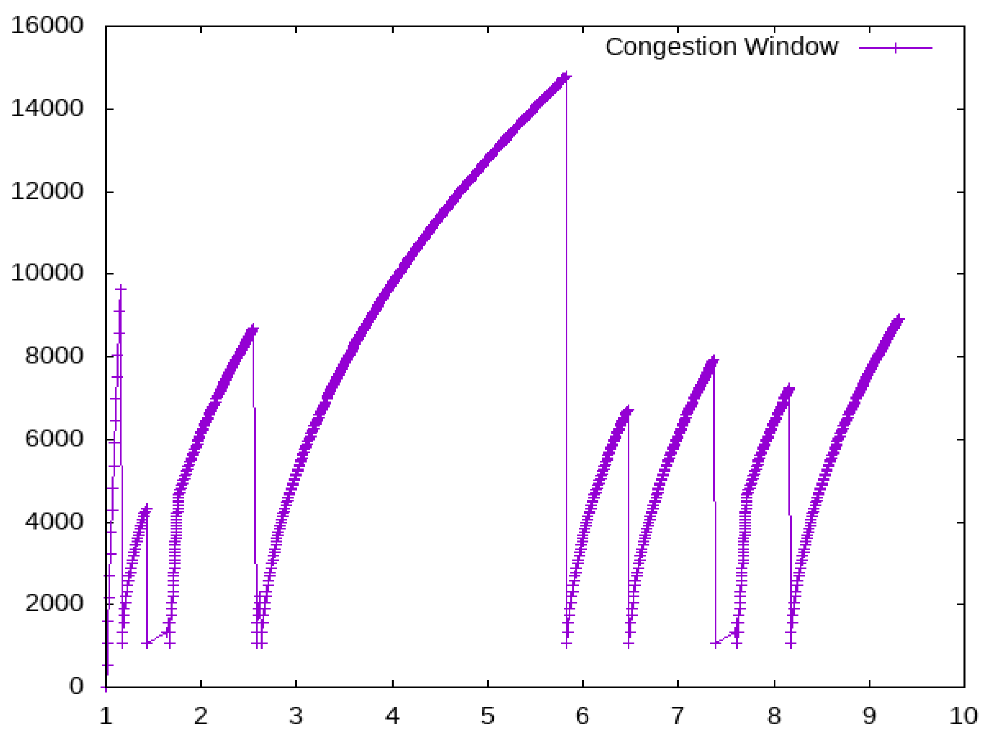
- Create a Gnuplot script file with the following lines:

```
set terminal png size 640,480
set output “cwnd.png”
plot “seventh.cwnd” using 1:2 title ‘Congestion Window’ with linespoints
exit
```

- Run the above script using Gnuplot:

```
$ gnuplot name-of-file.plt
```

- You should get the following graph inside the ns-3 directory (cwnd.png):



Congestion window size vs time in seconds

- The previous example was created with 2ms delay and 0.00001 channel error (or packet loss probability). The higher the error is, the more likely packets are dropped.
- Find where to modify these two parameters and reproduce the results under these settings:
    - Set the error to 0.00001:
        - Set the delay to 50ms.
        - Set the delay to 200ms.
        - Set the delay to 500ms.
        - Set the delay to 5s

    - Set the delay to 2ms:
        - Set error to 0.00002.
        - Set error to 0.00003.
        - Set error to 0.0002.
        - Set error to 0.02
        - Set error to 0.2

You may **need** to adjust the simulation time as the link delay increases.

Regarding the error model, you need to set ErrorUnit to "Packet" using the following line:

em->SetAttribute("ErrorUnit", StringValue ("ERROR_UNIT_PACKET"));

Bonus points (1 point): Figure out how to simulate and collect statistics (byte count) of average TCP throughput in bps and plot figures that show average TCP throughput vs link delay at a fixed error rate, TCP throughput vs error rate at a fixed link delay. This will help you write a better analysis of TCP performance below.

Report: Based on your simulation, your report should provide a detailed analysis on how TCP behaves under different operating conditions (as link delay, or loss changes). This is an open-ended question. You are welcome to experiment with other parameter settings in addition to those provided in the description above. Your task is to write a good report with evidence to back up your analysis.

Your report needs to be professional formatted. Figures need to be captioned, numbered, and cited in your report.

Grading

1. The total for lab 1 is 9+1 points;
3. All required lab tasks need to be completed; missing 1 task: -1;
4. Incomplete or partially correctly completed task: -0.5

5. Report should provide necessary figures and discussion to show understanding of the issues and labs performed. Lack of adequate figures or discussion: -0.5;
6. Figures must be captioned and numbered, AND cited in the discussion; otherwise -0.5